

APPLIED PHYSICS								
I B. TECH- I SEMESTER								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4BS08	BSC	L	T	P	C	CIE	SEE	Total
		3	1	0	4	30	70	100
COURSE OBJECTIVES: The course should enable the students to: - Learn the behaviour of matter waves and applications of Schrodinger wave equations in periodic potential energy of electron. - Understand the formation of energy bands in solids. - Gain the knowledge of carrier concentration and recombination process of semiconductor materials. - Learn the basic principles of laser and optical fibre. - Understand the development of nano technology and synthesis of nano materials by using different techniques.								
COURSE OUTCOMES: The student will able to: - Conclude the dual nature of material particles and able to explain how moving particles are associated with its energies - Analyse the energy bands in solids and accordingly classify the materials - Evaluate the mobility of charge carrier concentration of a given semiconductor material. - Justify how the graded index optical fiber is more efficient than step index optical fiber in fiber optic communication system. - Recommend appropriate synthesis method and explain the characterization techniques.								
UNIT-I	QUANTUM MECHANICS							
Introduction to quantum physics: Black body radiation, Planck's law, photoelectric effect, Compton effect, Waves and Particles, de Broglie Hypothesis, Matter Waves, Davisson and Germer's Experiment, Heisenberg's Uncertainty Principle, Schrodinger's Time Independent Wave Equation-Physical Significance of the wave Function, Particle in One Dimensional Potential Box.								
UNIT-II	INTRODUCTION TO ELECTRONIC MATERIALS							
Band theory - Free electron theory, Origin of Energy Band formation in Solids, Estimation of Fermi energy level, Density of states and energy band diagrams, Kronig-Penny model (to introduce origin of band gap), Energy bands in solids, E-k diagram, Direct and indirect bandgaps. Classification of Materials: Conductors, Semiconductors & Insulators, Effective mass of an Electron. Fermi-Dirac Statistics (Qualitative treatment).								
UNIT-III	SEMICONDUCTORS AND OPTOELECTRONICS							
Semiconductors: Intrinsic and extrinsic semiconductors, Dependence of Fermi level on carrier-concentration and temperature (equilibrium carrier statistics). Carrier generation and recombination, Carrier transport: diffusion and drift, Hall effect and its applications. Semiconductors design- PN junction-diode, Zener diode, fabrication and characterization techniques- Heterojunctions and associated band-diagrams. Optoelectronic devices: properties of photo detectors, solar cells, Semiconductor laser, Four-point probe measurement for carrier density.								
UNIT-IV	LASER & OPTICAL FIBER							
Laser: Characteristics of Laser beams, Energy levels in atoms, Einstein's theory of matter radiation interaction and A and B coefficients; amplification of light by population inversion, different types of lasers: gas lasers -He-Ne, solid-state lasers -ruby, Nd-YAG, Applications of Lasers. Fiber Optic Communication: Structure of Optical fibers, Basic principle of fibers, Acceptance angle and								

Numerical aperture, Types of Optical Fibers-Step Index and Graded Index fibers; Modes of fibers-SMSI, MMSI, MMGI., Optical fiber Communication System with block diagram. Applications of fibers, fiber optic sensors – Basic principle, Intrinsic, Extrinsic sensors. Working of Pressure and Temperature Sensors.		
UNIT-V	INTRODUCTION TO ENGINEERED MATERIALS	
Fundamentals of nano particles, nano scale, properties, Techniques for synthesis of nano materials -Sol-gel, Chemical vapour deposition (CVD) methods. Characterization of nanomaterials: Imaging methods- SEM, TEM, Scanning Probe Microscopy: STM. Fabrication method- quantum wire, Applications of Nano materials in engineering and Biomedical fields.		
TEXT BOOKS:		
1. Engineering Physics, B.K. Pandey, S. Chaturvedi – Cengage Learning 2. Haliday and Resnick, Physics – wiley 3. R. Robinett, “Quantum Mechanics”, OUP Oxford, 2006.IIndEdn. 4. P.K Palanisamy, Engineering Physics, Sitech Publications, 2013, IVthEdn. 5. Essentials of Nano Tecnology by Jeremy Ramsden.		
REFERENCE BOOKS:		
1. E. Hecht, “Optics”, Pearson Education, 2008. 2. J. Singh, Semiconductor Optoelectronics: Physics and Technology, Mc Graw-Hill inc.(1995) 3. R. Robinett, “Quantum Mechanics”, OUP Oxford, 2006. 4. O. 1Svelto, “Principles of Lasers”, Springer Science & Business Media, 2010. 5. D. A. Neamen, “Semiconductor Physics and Devices”, Times Mirror High EducationGroup, Chicago,1997.		
WEB REFERENCES:		
1. https://www.edx.org/course?search_query=semiconductor+physics 2. https://www.edx.org/course/nanotechnology-fundamentals-purdue-nano530x 3. https://www.edx.org/course/physics-electronic-polymers-pep-purdue-nano600		
E-TEXT BOOKS:		
1. http://www.phys.sinica.edu.tw/TIGP-NANO/Course/2010_Fall/classnotes/NanoB_week14.pdf 2. https://www.scribd.com/document/70908178/Semiconductor-Devices-Basic-Principles-Jaspri-Singh 3. https://www.scribd.com/doc/105174065/Fundamentals-of-Photonics 4. ftp://nozdr.ru/biblio/kolxo3/P/PE/PEo/Thyagarajan%20K.,%20Ghatak%20A.%20Lasers..%20Fundamentals%20and%20Applications%20(2ed.,%20GTP,%20Springer,%202010)(ISBN%20144196441X)(O)(674s)_PEo_.pdf 5. https://subodhtrpathi.files.wordpress.com/2012/01/optical-fiber-communications-by-gerd-keiser_2.pdf 6. http://www.hailienene.com/resources/nano-technology.pdf		
MOOC COURSE:		
1. http://nptel.ac.in/courses/115103030/(Four-point probe measurement for carrier density) 2. http://nptel.ac.in/courses/115102025/ (Fundamental concepts of semiconductors) 3. http://nptel.ac.in/courses/118104008/1 (Fundamentals of Nano technology) 4. http://nptel.ac.in/courses/118104008/13 (Nano structures, synthesis and characterization) 5. http://nptel.ac.in/courses/104104085/2(Lasers and its applications)		